apsara

Date: —/—/—

ACID Properties

ACID properties are rules that ensure reliable and consistent transaction processing in databases.

ACID stands for :

A → Atomicity

A transaction is treated as a single unit either all operations happen or none happen (all or nothing)

C → Consistency

A transaction must move database from one valid state to another valid state (maintains database validity)

I → Isolation

A transaction should always ~~transac~~ occurring in same database without any conflict (prevents transaction conflicts)

D → Durability

Once commited, data remains permanently saved even after system crash (ensures

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committed data is permanent)

TCL (Transaction Control Language)

TCL commands

1. START TRANSACTION

Begins a new transaction. changes can be committed or rolled back, subject to the database's transaction rules

Syntax: START TRANSACTION;

Ex:
```
START TRANSACTION;
UPDATE employee
SET salary = 30000
WHERE emp_id = 101;
```

2. COMMIT

Saves the changes made in the current transaction permanently

Syntax: COMMIT;

Ex:
```
START TRANSACTION;
UPDATE employee
SET salary = 30000
```


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WHERE emp_id = 101;
COMMIT;
The salary changes is committed to the database and normally cannot be undone using ROLLBACK

3. ROLLBACK
undoes uncommitted changes made in current transaction

Syntax: ROLLBACK;
Ex: START TRANSACTION;
UPDATE employee
SET salary = 30000
WHERE emp_id = 101;
ROLLBACK;
Salary returns to its previous value bez the update was rolled back.

ROLLBACK to SAVEPOINT syntax:
Synt: ROLLBACK TO SAVEPOINT savepoint_name;
This undoes changes made after a specified savepoint while keeping the transaction open

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4. SAVEPOINT

Creates a checkpoint inside a transaction so you can undo some changes without undoing the entire transaction

Syntax : SAVEPOINT savepoint_name;

Ex :
```
START TRANSACTION;
UPDATE employee
SET salary = 30000
WHERE emp_id = 101;
SAVEPOINT S1;
UPDATE employee
SET salary = 40000
WHERE emp_id = 102;
ROLLBACK TO SAVEPOINT S1;
COMMIT;
```

Result :
- employee 101 salary update is saved
- employee 102 salary update is undone
- the transaction is committed.

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DCL (Data Control Language)

DCL commands controls who can access a database and what actions they are allowed to perform

Ex : an administrator can allow one user to view a table and allow another user to update its data

DCL commands

1. GRANT

Grants permission to user

Syntax : Grant specific permission on a table

```
GRANT privileges
ON table_name
TO user;
```

Ex :
```
GRANT SELECT
ON employee
TO user1;
```

Result : user1 is granted permission to read data from the employee table.

other common permissions include : SELECT, INSERT UPDATE, DELETE

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syntax 2: Grant all privileges on all tables in a database

```
GRANT ALL PRIVILEGES
ON database_name.*
TO user;
```

ALL PRIVILEGES : means all privileges supported by the database system in that context.

database_name.* : refers to all tables in that database

To user: specifies the user receiving the permissions

2. REVOKE

Removes permissions that were previously granted to a user

syntax :
```
REVOKE privileges
ON table_name
FROM user;
```

Ex :
```
REVOKE SELECT
ON employees
FROM user1;
```

user1 loses the select permission on the employee table provided that permission was granted through the applicable grant.

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